



$$\begin{matrix} X_1 & X_2 \\ X_1 & \begin{bmatrix} 0.8 & 0.4 \\ 0.2 & 0.6 \end{bmatrix} = \begin{bmatrix} P(X_t = X_1 | X_{t-1} = X_1) & P(X_t = X_1 | X_{t-1} = X_2) \\ P(X_t = X_2 | X_{t-1} = X_1) & P(X_t = X_2 | X_{t-1} = X_2) \end{bmatrix} \end{matrix}$$

$$P(X_t = X_1) = P(X_{t-1} = X_1) P(X_t = X_1 | X_{t-1} = X_1) + P(X_{t-1} = X_2) P(X_t = X_1 | X_{t-1} = X_2)$$

$$P(X_t = X_2) = P(X_{t-1} = X_1) P(X_t = X_2 | X_{t-1} = X_1) + P(X_{t-1} = X_2) P(X_t = X_2 | X_{t-1} = X_2)$$

$$\begin{bmatrix} P(X_t = X_1) \\ P(X_t = X_2) \end{bmatrix} = \begin{bmatrix} P(X_t = X_1 | X_{t-1} = X_1) & P(X_t = X_1 | X_{t-1} = X_2) \\ P(X_t = X_2 | X_{t-1} = X_1) & P(X_t = X_2 | X_{t-1} = X_2) \end{bmatrix} \begin{bmatrix} P(X_{t-1} = X_1) \\ P(X_{t-1} = X_2) \end{bmatrix}$$

$$P(X_t) = A P(X_{t-1})$$

$$P(X_{t+1}) = A P(X_t)$$

$$P(X_{t+2}) = A P(X_{t+1}) = A A P(X_{t0})$$

$$P(X = X_1) = 0.8 P(X = X_1) + 0.4 P(X = X_2)$$

$$P(X = X_2) = 0.2 P(X = X_1) + 0.6 P(X = X_2)$$

$$P(X = X_1) = 2 P(X = X_2) \rightarrow P(X_2) = \frac{1}{2} P(X_1)$$

$$P(X_1) + P(X_2) = 1$$

$$P(X_1) + \frac{1}{2} P(X_1) = 1 \rightarrow P(X_1) = \frac{2}{3}$$

$$P(X_2) = \frac{1}{3}$$

- $S \rightarrow (s_1, \dots, s_n)$ estado $|S|$

- $A = \{a_1, \dots, a_n\}$

- Acciones: $S \rightarrow P(A)$

- $\text{acciones}(s) \subseteq A$

- Sucesor: $S \times A \rightarrow S$

- $\text{Sucesor}(s, a) \in S$

$\rightarrow \text{Sucesor}: S \times A \times S \rightarrow \mathbb{R} \quad \text{Sucesor}(s, a, s')$
 $\Pr[S_{t+1}=s' | S_t=s, A_t=a]$

- Costo-local: $S \times A \rightarrow \mathbb{R}^+$

- $\text{Costo-local}(s, a) \in \mathbb{R}^+$

$\rightarrow \text{recompensa}: S \times A \times S \rightarrow \mathbb{R} \quad \text{recompensa}(s, a, s')$

- terminal $S \rightarrow \{F, V\}$

- $\text{terminal}(s)$

MDP

$S = \{s_1, \dots, s_n\}$

$A = \{a_1, \dots, a_m\}$

Alegal: $S \rightarrow P(A)$

$T: S \times A \times S \rightarrow \mathbb{R}$

$r: S \times A \times S \rightarrow \mathbb{R}$

$s_t \in S$

$0 \leq \gamma \leq 1$

¿Qué quiero?

Para cada $s \in S$

$\mathbb{E}[R_t | S_t=s]$ son máximo

$R_t = r_t + \gamma r_{t+1} + \gamma^2 r_{t+2} + \dots$

¿Cómo lo consigo?

$\pi: S \rightarrow A$

$\Pi: S \times A \rightarrow \mathbb{R}$

¿Y si tengo 2 Políticas π_1 y π_2 , Cual es mejor?

Calculamos la Función de Valor de Estado de π

$$V^\pi(s) = E^\pi[R_t | S_t = s]$$

$$V^\pi(s) = \sum_{s' \in S} T(s, \pi(s), s') [r(s, \pi(s), s') + \gamma E[V^\pi(s') | S_{t+1} = s']]$$

$$V^\pi(s) = \sum_{a \in A_{\text{legal}}(s)} \pi(s, a) \sum_{s' \in S} T(s, a, s') [r(s, a, s') + \gamma V^\pi(s')]$$

Si:
 $V^{\pi_1}(s) \leq V^{\pi_2}(s) \quad \forall s \in S$
 entonces $\pi_1 \leq \pi_2$

π_1 y π_2
 y hago
 π_3 tal que $\pi_3(s) = \begin{cases} \pi_1(s) & \text{si } V^{\pi_1}(s) > V^{\pi_2}(s) \\ \pi_2(s) & \text{si } V^{\pi_2}(s) \geq V^{\pi_1}(s) \end{cases}$

entonces
 $\pi_1 \leq \pi_3$ y $\pi_2 \leq \pi_3$
 entonces $\exists \pi^*$ tal que
 $V^{\pi^*}(s) \geq V^\pi(s) \quad \forall \pi$

$$\begin{bmatrix} V^\pi(s_1) \\ V^\pi(s_2) \\ \vdots \\ V^\pi(s_n) \end{bmatrix} = \begin{bmatrix} \sum_{s' \in S} T(s_1, \pi(s_1), s') [r(s_1, \pi(s_1), s') + \gamma V^\pi(s')] \\ \sum_{s' \in S} T(s_2, \pi(s_2), s') [r(s_2, \pi(s_2), s') + \gamma V^\pi(s')] \\ \vdots \\ \sum_{s' \in S} T(s_n, \pi(s_n), s') [r(s_n, \pi(s_n), s') + \gamma V^\pi(s')] \end{bmatrix}$$

$$V^\pi = f(V^\pi)$$

Función de valor Estado-Acción
 $Q^\pi(s, a) = E^\pi[R_t | S_t = s, A_t = a]$

$$Q^\pi(s, a) = \sum_{s' \in S} T(s, a, s') [r(s, a, s') + \gamma V^\pi(s')] \\ = \sum_{a' \in A(s')} \pi(s', a') Q^\pi(s', a')$$


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def calcula_V(mdp, Politica, tolerancia = 1e-6):
    V = {s: 0 for s in mdp.estados()}
    for _ in range(100):
        delta = 0
        for s in mdp.estados():
            Vp = sum(mdp.sucesor(s, Politica[s], s_p) * [mdp.recompensa(s, Politica[s], s_p) + mdp.descuento * V[s_p]
                    for s_p in mdp.estados()])
            delta = max(delta, abs(Vp - V[s]))
            V[s] = Vp
        if delta < tolerancia:
            break
    return V

```

s_1	s_2
s_3	s_4

$$\pi_1(s, \rightarrow) = 1 \quad \forall s$$

$$\pi_2(s, a) = \frac{1}{4} \quad \forall s, \forall a$$

↑ acción aleatoria

$$V^{\pi_1}(s_1) = P(s_1, \rightarrow, s_1) [-1 + \gamma V^{\pi_1}(s_1)] + P(s_1, \rightarrow, s_2) [1] + P(s_1, \rightarrow, s_3) [-1 + \gamma V^{\pi_1}(s_3)] + P(s_1, \rightarrow, s_4) [-1 + \gamma V^{\pi_1}(s_4)]$$

$$V^{\pi_1}(s_1) = 0.05 [-1 + \gamma V^{\pi_1}(s_1)] + 0.9 + 0.05 [-1 + \gamma V^{\pi_1}(s_3)]$$

$$V^{\pi_1}(s_3) = 0.05 [-1 + \gamma V^{\pi_1}(s_1)] + 0.9 + 0.05 [-1 + \gamma V^{\pi_1}(s_3)]$$

$$V^{\pi_1}(s_4) = 0.9 [-1 + \gamma V^{\pi_1}(s_4)] + 0.05 [-1 + \gamma V^{\pi_1}(s_4)] + 0.05 [1]$$

$$\pi_1 < \pi_2 \quad \forall s$$

$$V^{\pi_1}(s) \leq V^{\pi_2}(s) \quad \forall s$$

$$\pi_1 \neq \pi_2$$

$$\pi_3 \text{ tal que } \pi_3(s) = \begin{cases} \pi_1(s) & \text{si } V^{\pi_1}(s) \geq V^{\pi_2}(s) \\ \pi_2(s) & \text{o.c.} \end{cases} \quad \forall s \in S$$

$$\pi_1 \leq \pi_3$$

$$\pi_2 \leq \pi_3$$

π^* es una política óptima si

$$V^{\pi^*}(s) \geq V^{\pi}(s) \quad \forall \pi$$

Función de valor estado-acción

$$Q^{\pi}(s, a) = E^{\pi}[R_t | S_t = s, A_t = a]$$

$$Q^{\pi}(s, a) = \sum_{s' \in S} T(s, a, s') [r(s, a, s') + \gamma V^{\pi}(s')]$$

$$V^{\pi}(s) = \sum_{a \in A(s)} \pi(s, a) Q^{\pi}(s, a)$$

$$\pi^* \rightarrow V^* Q^*$$

$$\pi^*(s, a) = \begin{cases} 1 & \text{Para } a^* \neq a \\ 0 & \text{o.c.} \end{cases} \quad \text{donde } a^* = \arg \max_a Q^*(s, a)$$

$$V^*(s) = \max_a Q^*(s, a)$$

Equación de optimalidad de Bellman

$$V^*(s) = \max_a \left[\sum_{s' \in S} T(s, a, s') [r(s, a, s') + \gamma V^*(s')] \right]$$